

Beacon

Self-hosted uptime and status platform

Independent of the internship Library app. One product from CS through Kubernetes.

What it is

You run Beacon. It watches websites and APIs. If they fail, it opens an incident and can alert you. The public sees a status page (operational / degraded / down). You see a control panel (login, monitors, history).

Pieces you will build

api - users, monitors, incidents, history. probe - worker that checks HTTP/TCP/DNS. alerter - webhook (later email). web-public - status page, no login. web-admin - login and CRUD. postgres - source of truth. redis - job queue. edge - nginx or Ingress later. You do not start with all of these; you grow into them.

How a check works

1. Save a monitor (URL, interval, HTTP GET). 2. API enqueues a job on Redis. 3. Probe dequeues, resolves DNS, connects, records latency and status. 4. If up flips to down, an incident opens and alerter fires. 5. Public status page reads latest state (no secrets).

Data

users; monitors (name, type, target, interval); results (when, ok, latency_ms, error); incidents; api_keys (hashed).

Stack

API + probe + alerter: Python (FastAPI) or Go - pick one and stay. PostgreSQL. Redis. Small separate UI (or server-rendered first). Git from day one.

Phase by phase - what you build

Phase 0 CS: CLI beacon-check - one URL, ok/fail, write a result line to a file, exit code. No Docker, no Kubernetes.

Phase 1 Network: CLI grows - HTTP vs TCP vs DNS, resolved IP, port, TLS cert expiry, timeouts.

Phase 2 Cybersecurity: tiny API - register/login, hashed passwords, hashed API keys, never log tokens.

Phase 3 System administration: Postgres + Redis + API as systemd services, nginx, logs, Postgres backup, SSH-only admin.

Phase 4 Cloud: same on one cloud VM, security group (80/443 public, SSH yours), object storage for logo, know the bill.

Phase 5 DevOps: Dockerfiles, Compose for the full stack, CI test + image build.

Phase 6 Kubernetes: Deployments, Services, Ingress, Secrets/ConfigMaps, probe replicas, Helm, GitOps, NetworkPolicy.

Later

Not now: mobile app, operators, service mesh, multi-region probes.

Beacon - project brief

Current heading

Phase 0 - CLI check. Closed until you say we start.